

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Junior Backend Developer | Java & Spring Boot

REST APIs ▪ Event-Driven Microservices ▪ Hexagonal Architecture

Software engineer specializing in the Java ecosystem, building REST APIs and event-driven microservices with clean, testable architecture. Hands-on Spring Boot 3 work includes Kafka-based microservices (outbox pattern, dead-letter topics, idempotent processing), REST APIs with JPA and Jakarta Validation, and integration testing with Testcontainers. Currently deepening my Java skills through self-study of hexagonal architecture (ports & adapters) with Spring Boot and architectural design patterns. Also proficient in Golang, TypeScript and React, with professional R&D experience building LLM-based agents at ITI. Master's Degree in Computer Engineering (final grade 8.8/10) and certified B2 English.

Core Competencies

- **Java Backend**: Java, Spring Boot 3, Spring Data JPA, Jakarta Validation, REST APIs, DTOs/mappers, centralized error handling, JUnit, Testcontainers
- **Architecture**: Hexagonal architecture (ports & adapters), event-driven microservices, outbox pattern, dead-letter topics, idempotent consumers
- **Messaging & Infrastructure**: Apache Kafka, NATS, Docker, Docker Compose, Apache APISIX, PostgreSQL
- **Other Languages & Frontend**: Golang, TypeScript/JavaScript, React, Python, Kotlin, SQL
- **LLM/AI Engineering**: LangChain, LangGraph, Gemini API, OpenAI API, RAG pipelines
- **Tools & Practices**: Git, GitHub Actions, TDD, OAuth 2.0/JWT, Firebase, Supabase, Vercel, Render, NeoVim

Professional Experience

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
R&D Developer — *Feb 2026–Jun 2026*

- Implemented LLM-based improvements to a search engine
- Built a LangGraph multi-step agent that converts natural-language questions into validated, read-only SQL queries to resolve data-access policies over a PostgreSQL data-governance schema
- Applied safety-conscious agentic design: schema/table selection, structured SQL generation, execution validation, retry-on-error
- Worked within an application focused on management, governance, and sovereignty of industrial data

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
Full-Stack Developer Intern — *Feb 2024–Jun 2024*

- Developed and maintained the corporate personnel management application (Intranet)
- Implemented new frontend features using React and TypeScript, improving internal usability
- Contributed to backend development, ensuring smooth integration between services

Selected Java Projects

- **Kafka + Spring Boot Ticketing Platform** – *Java, Spring Boot 3, Apache Kafka, Testcontainers*: Event-driven microservices (reservation, payment, notification, audit) using the outbox pattern, dead-letter topics, idempotent processing, and integration tests with Testcontainers.
- **English Learning API** – *Java, Spring Boot, Spring Data JPA, Jakarta Validation*: REST API for practicing English phrasal verbs and collocations. JPA entities, DTOs and mappers, request validation, centralized error handling, and randomized practice challenges with attempt tracking.

- **HTTP Server from TCP (Java)** – *Java, JUnit*: HTTP/1.1 server implemented directly on raw sockets with no framework, built to deepen Java fundamentals: I/O streams, concurrency with executors, and unit testing with JUnit.

Other Projects

- **FaaS with Golang** – *Go, Docker, NATS JetStream, Apache APISIX*: Distributed Function-as-a-Service platform where workers execute arbitrary Docker-image functions. NATS JetStream work-queues, APISIX gateway for JWT auth and rate limiting, horizontal scaling via Docker Compose replicas.
- **AIDucate** – *Go, Gin, PostgreSQL, React, TypeScript, Gemini API*: AI-powered SaaS platform generating personalized study plans and exercises. JWT-authenticated backend with PostgreSQL; React frontend on Vercel, backend on Render. *(In progress)*
- **ValenMove** – *Kotlin, MVVM, JSoup*: Native Android app unifying Valencia's public transport (EMT, Metrovalencia, Valenbisi) with real-time arrivals from reverse-engineered public APIs and Open Data. Final Degree Project, grade 9/10.

Continuous Learning

- **Hexagonal Architecture in Java with Spring Boot** – self-study (in progress): ports & adapters, framework-independent domain, Spring Boot adapters.
- **Architectural Design Patterns** – self-study (in progress): software architecture patterns and their trade-offs.

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  Completed (Sep 2026), final grade 8.8/10. Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis (TFG): "ValenMove – Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (certified B2, confident conversational).

References

- Available upon request.